211120: altered breccia, Deflector prospect

(Mougooderra Formation, Murchison Supergroup, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Yalgoo (SH 50-2), Mellenbye (2140)
MGA Zone 50, 438968E 6828203N

Sampled on 20 May 2019

This sample was collected from the 110.0 – 110.5 m depth interval of drillcore BDRCD075, drilled in 2006 by Batavia Mining Ltd at their Deflector prospect (Cobb et al., 2006). The drillhole is located about 8.9 km south-southeast of Five Mile Well, 7.4 km east-northeast of Yallabyne Well, and 0.3 km southwest of the Deflector West open pit.

Geological context

The unit sampled is the Mougooderra Formation of the Murchison Supergroup (Van Kranendonk et al., 2013; Zibra, 2021). The Mougooderra Formation is a dominantly siliciclastic unit that unconformably overlies the Polelle and Norie Groups, and is intruded by granitic rocks of the 2650–2600 Ma Bald Rock Supersuite. A maximum depositional age of 2758 ± 4 Ma was determined for a volcaniclastic mudstone of the formation (GSWA 211101, Lu et al., 2016), collected about 57 km to the northeast, and an imprecise younger limit for deposition is provided by a Lu–Hf isochron date of 2686 ± 18 Ma for metamorphic garnet in a pelitic rock located 105 km to the southeast (GSWA 207692, Parmenter, 2019; Parmenter et al., 2020). The present sample is a poorly sorted, polymictic breccia (Fig. 1) containing about 75% felsic volcanic clasts 5–50 mm in size, and 25% biotite-altered matrix.

Petrographic description

The sample is a hydrothermally altered rock, consisting of about 50–55% biotite, 30–35% feldspar, 7% clinozoisite, 5% quartz, 3% carbonate minerals, and zircon. Colourless to pale brown biotite forms non-aligned, decussate aggregates of subhedral laths up to 0.2 mm long. Feldspar forms round to oval grains up to 2 mm in diameter, which have a radial spherulitic texture, and overprinting by very fine-grained biotite precludes precise identification. Clinozoisite crystals are subhedral, colourless, up to 0.8 mm in size, and appear to be essentially contemporaneous with biotite. Clinozoisite also forms veins up to 1 mm wide. Quartz occurs in a 2 mm-wide vein with a microcrystalline polygonal texture. The vein also contains minor carbonate and there are other small carbonate veins or infills that lack quartz. The rock is not obviously deformed and the mineralogy is interpreted as hydrothermal.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown, and mainly subhedral to euhedral. The crystals are up to 300 µm long, and equant to elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and a few crystals appear to contain older cores. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 12–13 and 18–19 July 2019, using SHRIMP-B. Analyses 1.1 to 61.1 (spot numbers 1–61) were obtained during the first session, together with 13 analyses of the CZ3 standard, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.61% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.26% (1σ). Analyses 62.1 to 94.1 (spot numbers 62–95) were obtained during the second session, together with nine analyses of the CZ3 standard, of which seven analyses indicated an external spot-to-spot (reproducibility) uncertainty of 1.04% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.44% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the second session was corrected by reference to the OGC1 standard; measured ratios were increased by 0.24%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Ninety-five analyses were obtained from 94 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3) and a probability diagram (Fig. 4).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Twenty-three analyses are >5% discordant and one imprecise analysis was obtained during a brief period of instrument instability. The dates obtained from these 24 analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 71 analyses can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

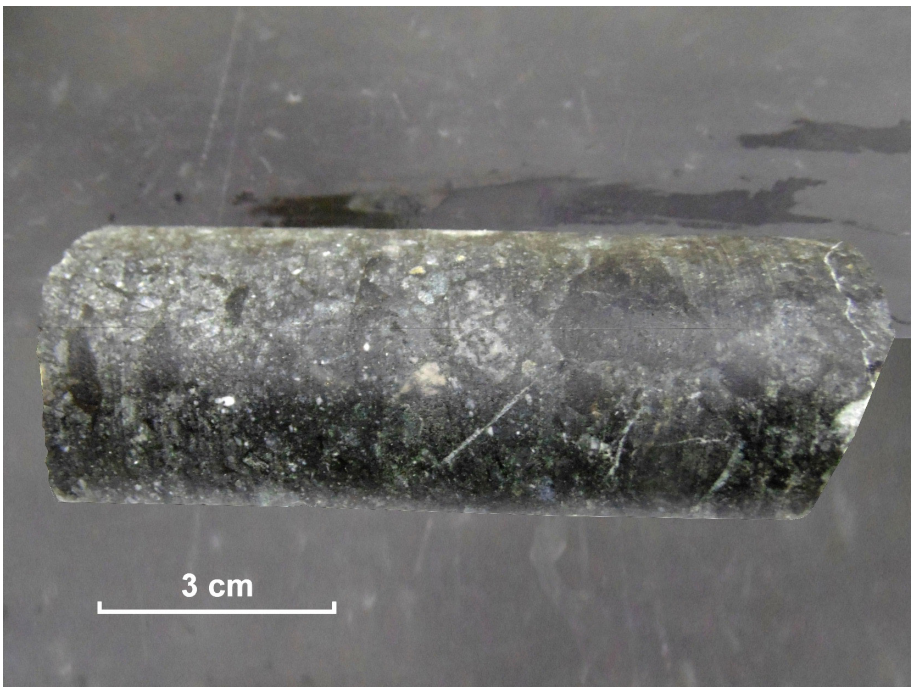

Figure 1. **Drillcore image of sample 211120: altered breccia, Deflector prospect**

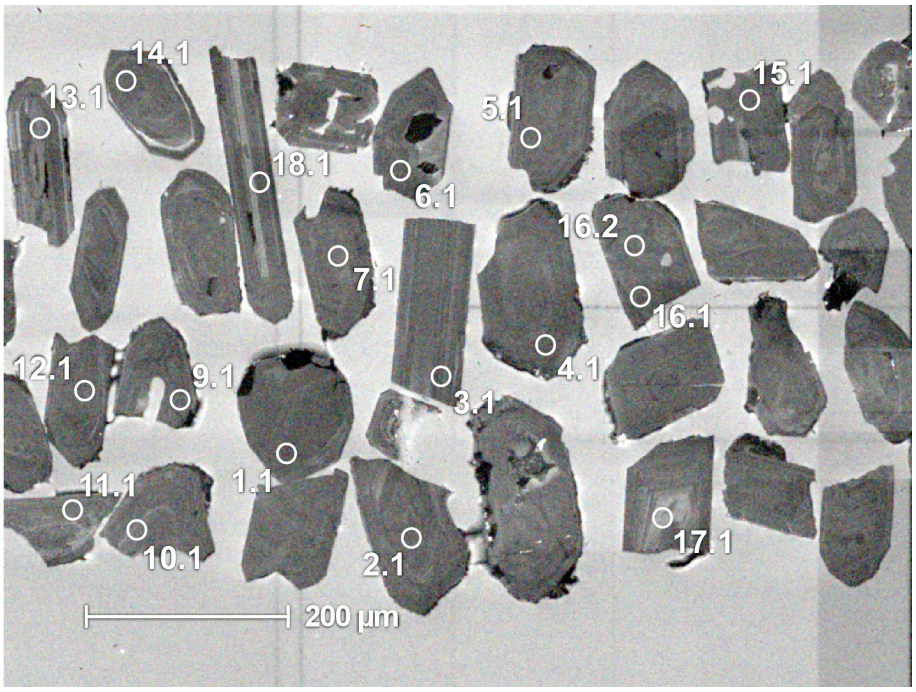

Figure 2. **Cathodoluminescence image of representative zircons from sample 211120: altered breccia, Deflector prospect. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 211120: altered breccia, Deflector prospect

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$	± 1σ	$^{207}Pb/^{206}Pb$	± 1σ	$^{238}U/^{206}Pb^*$	± 1σ	$^{207}Pb^*/^{206}Pb^*$	± 1σ	$^{238}U/^{206}Pb^*$ date (Ma)	±1σ	$^{207}Pb^*/^{206}Pb^*$ date (Ma)	±1σ	Disc. (%)
Y	16	16.1	1454	339	0.23	0.017	2.072	0.023	0.17691	0.00030	2.072	0.023	0.17675	0.00031	2539	24	2623	3	3.2
Y	91	16.2	2209	668	0.30	0.021	2.070	0.024	0.17777	0.00042	2.071	0.024	0.17759	0.00042	2540	25	2630	4	3.4
S	53	53.1	1053	286	0.27	0.018	2.040	0.015	0.17790	0.00033	2.040	0.015	0.17774	0.00034	2571	16	2632	3	2.3
S	35	35.1	894	158	0.18	0.024	1.987	0.015	0.18163	0.00060	1.988	0.015	0.18141	0.00061	2627	17	2666	6	1.4
S	31	31.1	780	180	0.23	0.022	1.988	0.018	0.18221	0.00039	1.988	0.018	0.18201	0.00039	2627	20	2671	4	1.7
S	83	83.1	1103	496	0.45	0.004	1.942	0.023	0.18208	0.00047	1.943	0.023	0.18205	0.00047	2677	27	2672	4	-0.2
S	20	20.1	879	951	1.08	0.010	2.026	0.016	0.18376	0.00080	2.026	0.016	0.18367	0.00081	2586	17	2686	7	3.7
S	2	2.1	1164	762	0.66	0.009	1.969	0.015	0.18443	0.00035	1.969	0.015	0.18435	0.00035	2647	17	2692	3	1.7
S	29	29.1	845	212	0.25	0.008	2.034	0.016	0.18525	0.00043	2.035	0.016	0.18518	0.00044	2577	17	2700	4	4.5
S	72	72.1	472	264	0.56	0.014	2.018	0.025	0.18534	0.00061	2.018	0.025	0.18522	0.00062	2595	27	2700	5	3.9
S	63	63.1	714	179	0.25	0.032	1.962	0.024	0.18560	0.00056	1.962	0.024	0.18532	0.00056	2655	27	2701	5	1.7
S	57	57.1	907	456	0.50	0.025	1.988	0.021	0.18568	0.00039	1.988	0.021	0.18546	0.00039	2627	23	2702	4	2.8
S	58	58.1	659	474	0.72	0.012	1.975	0.015	0.18691	0.00043	1.975	0.015	0.18681	0.00044	2641	17	2714	4	2.7
S	82	82.1	374	81	0.22	0.086	2.025	0.025	0.18830	0.00067	2.027	0.025	0.18754	0.00069	2586	27	2721	6	5.0
S	27	27.1	484	227	0.47	0.017	1.997	0.017	0.18826	0.00052	1.998	0.017	0.18811	0.00052	2616	18	2726	5	4.0
S	65	65.1	905	299	0.33	0.012	1.873	0.022	0.18823	0.00051	1.873	0.022	0.18812	0.00051	2757	28	2726	4	-1.2
S	70	70.1	649	162	0.25	0.023	1.946	0.024	0.18849	0.00057	1.946	0.024	0.18828	0.00058	2673	27	2727	5	2.0
S	81	81.1	641	202	0.32	0.021	1.951	0.024	0.18849	0.00080	1.951	0.024	0.18830	0.00081	2668	27	2727	7	2.2
S	7	7.1	736	214	0.29	0.015	1.953	0.015	0.18864	0.00047	1.953	0.015	0.18851	0.00047	2665	18	2729	4	2.3
S	6	6.1	475	78	0.16	0.064	1.920	0.016	0.18910	0.00056	1.921	0.016	0.18853	0.00058	2702	19	2729	5	1.0
S	8	8.1	489	262	0.54	0.017	1.961	0.016	0.18879	0.00052	1.961	0.016	0.18864	0.00052	2656	18	2730	5	2.7
S	36	36.1	667	176	0.26	0.017	1.981	0.016	0.18892	0.00044	1.981	0.016	0.18877	0.00044	2634	18	2731	4	3.6
S	17	17.1	159	49	0.31	0.065	1.887	0.020	0.18950	0.00092	1.888	0.020	0.18892	0.00095	2740	24	2733	8	-0.3
S	93	92.1	607	197	0.32	0.016	1.847	0.023	0.18909	0.00061	1.847	0.023	0.18895	0.00061	2789	29	2733	5	-2.1
S	46	46.1	801	202	0.25	0.010	1.950	0.015	0.18919	0.00040	1.950	0.015	0.18910	0.00040	2668	17	2734	3	2.4
S	32	32.1	663	289	0.44	0.023	1.947	0.015	0.18964	0.00045	1.948	0.015	0.18944	0.00046	2672	18	2737	4	2.4
S	5	5.1	780	184	0.24	0.016	1.964	0.015	0.18964	0.00043	1.964	0.015	0.18950	0.00043	2653	17	2738	4	3.1
S	19	19.1	821	232	0.28	-0.002	1.882	0.015	0.18985	0.00039	1.882	0.015	0.18987	0.00040	2747	18	2741	3	-0.2
S	48	48.1	721	222	0.31	0.022	1.935	0.020	0.19010	0.00043	1.935	0.020	0.18990	0.00043	2685	23	2741	4	2.1
S	88	88.1	427	302	0.71	0.016	1.909	0.024	0.19060	0.00071	1.909	0.024	0.19046	0.00072	2715	28	2746	6	1.1
S	73	73.1	690	148	0.21	0.035	1.947	0.023	0.19089	0.00054	1.948	0.023	0.19057	0.00054	2671	27	2747	5	2.8

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	13	13.1	264	63	0.24	0.243	1.949	0.018	0.19279	0.00074	1.954	0.018	0.19062	0.00085	2664	21	2748	7	3.0
S	28	28.1	554	683	1.23	0.030	1.905	0.016	0.19090	0.00049	1.906	0.016	0.19063	0.00050	2719	19	2748	4	1.0
S	60	60.1	585	219	0.37	0.038	1.963	0.016	0.19117	0.00047	1.964	0.016	0.19082	0.00048	2653	18	2749	4	3.5
S	47	47.1	1143	168	0.15	0.005	1.931	0.014	0.19092	0.00059	1.931	0.014	0.19087	0.00059	2691	17	2750	5	2.2
S	23	23.1	748	147	0.20	0.015	1.932	0.015	0.19114	0.00047	1.932	0.015	0.19100	0.00047	2689	18	2751	4	2.2
S	4	4.1	597	323	0.54	0.057	1.965	0.016	0.19179	0.00048	1.966	0.016	0.19128	0.00049	2651	18	2753	4	3.7
S	24	24.1	627	167	0.27	0.024	1.926	0.015	0.19155	0.00088	1.926	0.015	0.19134	0.00088	2696	18	2754	8	2.1
S	11	11.1	275	75	0.27	0.031	1.957	0.018	0.19161	0.00064	1.958	0.018	0.19133	0.00065	2660	20	2754	6	3.4
S	41	41.1	709	141	0.20	0.005	1.933	0.015	0.19156	0.00044	1.933	0.015	0.19152	0.00044	2688	18	2755	4	2.4
S	49	49.1	600	475	0.79	0.017	1.944	0.016	0.19175	0.00048	1.945	0.016	0.19160	0.00048	2675	18	2756	4	2.9
S	3	3.1	419	161	0.38	0.032	1.900	0.016	0.19195	0.00057	1.901	0.016	0.19166	0.00058	2725	20	2757	5	1.2
S	92	91.1	457	136	0.30	0.004	1.964	0.024	0.19183	0.00064	1.964	0.024	0.19179	0.00064	2653	28	2758	5	3.8
S	37	37.1	285	148	0.52	0.033	1.944	0.018	0.19218	0.00066	1.945	0.018	0.19189	0.00067	2675	20	2758	6	3.0
S	40	40.1	252	90	0.36	0.027	1.974	0.019	0.19214	0.00082	1.974	0.019	0.19190	0.00082	2642	21	2759	7	4.2
S	25	25.1	489	120	0.25	0.043	1.948	0.016	0.19229	0.00054	1.949	0.016	0.19191	0.00055	2670	19	2759	5	3.2
S	80	80.1	219	71	0.33	0.022	1.953	0.026	0.19222	0.00083	1.953	0.026	0.19203	0.00084	2665	30	2760	7	3.4
S	34	34.1	333	168	0.51	0.016	1.957	0.017	0.19218	0.00065	1.957	0.017	0.19204	0.00065	2661	20	2760	6	3.6
S	42	42.1	321	117	0.37	0.005	1.885	0.017	0.19212	0.00123	1.885	0.017	0.19208	0.00123	2743	20	2760	11	0.6
S	50	50.1	483	212	0.44	0.004	1.958	0.016	0.19217	0.00055	1.958	0.016	0.19213	0.00055	2660	19	2761	5	3.7
S	94	93.1	496	121	0.24	0.006	1.907	0.023	0.19220	0.00060	1.907	0.023	0.19214	0.00060	2718	28	2761	5	1.5
S	45	45.1	217	134	0.62	-0.016	1.917	0.019	0.19200	0.00081	1.917	0.019	0.19214	0.00081	2706	22	2761	7	2.0
S	18	18.1	438	131	0.30	0.019	1.935	0.016	0.19232	0.00055	1.936	0.016	0.19215	0.00056	2685	19	2761	5	2.8
S	62	62.1	519	127	0.24	0.029	1.888	0.023	0.19248	0.00062	1.889	0.023	0.19222	0.00062	2739	28	2761	5	0.8
S	79	79.1	406	234	0.58	0.014	1.899	0.024	0.19238	0.00063	1.899	0.024	0.19226	0.00063	2727	28	2762	5	1.2
S	56	56.1	241	98	0.41	0.041	1.933	0.018	0.19283	0.00074	1.933	0.018	0.19247	0.00076	2688	21	2763	6	2.7
S	10	10.1	357	76	0.21	0.037	1.880	0.017	0.19316	0.00061	1.880	0.017	0.19284	0.00062	2749	20	2767	5	0.6
S	86	86.1	262	87	0.33	0.014	1.962	0.026	0.19301	0.00083	1.962	0.026	0.19288	0.00083	2655	29	2767	7	4.0
S	87	87.1	514	129	0.25	0.000	1.895	0.023	0.19291	0.00061	1.895	0.023	0.19291	0.00061	2732	28	2767	5	1.3
S	54	54.1	268	121	0.45	0.057	1.888	0.024	0.19379	0.00079	1.889	0.024	0.19329	0.00080	2738	29	2770	7	1.2
S	30	30.1	142	107	0.76	-0.021	1.877	0.020	0.19350	0.00091	1.876	0.020	0.19369	0.00092	2754	25	2774	8	0.7
S	15	15.1	715	352	0.49	0.010	1.892	0.015	0.19453	0.00045	1.892	0.015	0.19445	0.00045	2735	18	2780	4	1.6
S	14	14.1	955	160	0.17	0.031	1.896	0.014	0.19922	0.00038	1.897	0.014	0.19894	0.00038	2730	17	2818	3	3.1

Table 1. continued

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
S	43	43.1	579	187	0.32	0.002	1.787	0.014	0.21242	0.00079	1.787	0.014	0.21240	0.00079	2865	19	2924	6	2.0
S	76	76.1	366	129	0.35	0.037	1.773	0.022	0.21275	0.00071	1.774	0.022	0.21242	0.00072	2882	30	2924	6	1.4
S	77	77.1	267	8	0.03	0.026	1.796	0.023	0.21520	0.00079	1.796	0.023	0.21497	0.00080	2853	31	2943	6	3.1
S	89	89.1	227	67	0.29	0.023	1.769	0.023	0.21730	0.00083	1.769	0.023	0.21709	0.00084	2888	32	2959	6	2.4
S	59	59.1	296	177	0.60	0.033	1.763	0.016	0.21822	0.00068	1.763	0.016	0.21793	0.00069	2896	21	2966	5	2.3
S	1	1.1	604	194	0.32	0.016	1.767	0.014	0.21853	0.00048	1.768	0.014	0.21839	0.00048	2890	19	2969	4	2.6
S	51	51.1	483	343	0.71	0.040	1.797	0.015	0.22120	0.00055	1.797	0.015	0.22084	0.00056	2852	19	2987	4	4.5
S	22	22.1	482	12	0.02	0.073	1.670	0.020	0.23570	0.00061	1.671	0.020	0.23506	0.00062	3023	29	3087	4	2.1
D	66	66.1	1846	876	0.47	0.038	3.163	0.037	0.14895	0.00045	3.164	0.037	0.14861	0.00046	1770	18	2330	5	24.0
D	67	67.1	2100	2916	1.39	0.027	2.865	0.033	0.14958	0.00071	2.866	0.033	0.14934	0.00071	1930	20	2338	8	17.5
D	71	71.1	1280	547	0.43	0.103	2.454	0.029	0.16461	0.00107	2.457	0.029	0.16369	0.00108	2201	23	2494	11	11.7
D	38	38.1	871	198	0.23	0.106	2.469	0.019	0.16963	0.00046	2.472	0.019	0.16868	0.00049	2190	15	2545	5	13.9
D	95	94.1	1449	2359	1.63	0.023	2.379	0.031	0.17018	0.00046	2.379	0.031	0.16997	0.00046	2262	25	2557	5	11.5
D	39	39.1	2049	810	0.40	0.024	2.336	0.017	0.17079	0.00068	2.337	0.017	0.17057	0.00068	2297	14	2563	7	10.4
D	26	26.1	1399	376	0.27	0.029	2.343	0.017	0.17312	0.00031	2.344	0.017	0.17286	0.00032	2291	15	2586	3	11.4
D	74	74.1	1129	417	0.37	0.027	2.439	0.029	0.17607	0.00050	2.440	0.029	0.17583	0.00050	2215	23	2614	5	15.3
D	75	75.1	1099	563	0.51	0.043	2.246	0.027	0.17718	0.00050	2.247	0.027	0.17679	0.00051	2373	24	2623	5	9.5
D	44	44.1	709	163	0.23	0.074	2.307	0.018	0.17804	0.00072	2.308	0.018	0.17738	0.00074	2320	16	2629	7	11.7
D	61	61.1	1002	224	0.22	0.022	2.146	0.016	0.18097	0.00060	2.147	0.016	0.18078	0.00061	2465	16	2660	6	7.3
D	78	78.1	757	161	0.21	0.034	2.107	0.025	0.18238	0.00054	2.108	0.025	0.18208	0.00054	2503	26	2672	5	6.3
D	21	21.1	1694	720	0.42	0.027	2.069	0.206	0.18251	0.01105	2.070	0.206	0.18227	0.01105	2541	209	2674	100	5.0
D	69	69.1	939	225	0.24	0.024	2.174	0.026	0.18312	0.00053	2.174	0.026	0.18290	0.00053	2439	25	2679	5	9.0
D	33	33.1	216	83	0.39	0.040	2.157	0.021	0.18342	0.00077	2.158	0.021	0.18307	0.00078	2455	20	2681	7	8.4
D	64	64.1	735	262	0.36	0.015	2.063	0.025	0.18636	0.00057	2.063	0.025	0.18623	0.00057	2548	26	2709	5	6.0
D	85	85.1	769	143	0.19	0.031	2.168	0.026	0.18805	0.00054	2.168	0.026	0.18778	0.00055	2445	25	2723	5	10.2
D	55	55.1	740	188	0.25	0.036	2.057	0.016	0.18900	0.00044	2.058	0.016	0.18868	0.00045	2553	17	2731	4	6.5
D	90	90.1	307	99	0.32	0.085	2.206	0.029	0.19064	0.00086	2.208	0.029	0.18988	0.00089	2408	27	2741	8	12.1
D	9	9.1	495	69	0.14	0.029	2.057	0.017	0.19038	0.00054	2.058	0.017	0.19012	0.00055	2553	18	2743	5	6.9
D	52	52.1	415	118	0.28	0.008	2.022	0.017	0.19217	0.00058	2.022	0.017	0.19209	0.00059	2590	18	2760	5	6.2
D	84	84.1	888	460	0.52	0.014	1.986	0.024	0.20091	0.00052	1.986	0.024	0.20079	0.00053	2629	27	2833	4	7.2
D	68	68.1	306	316	1.03	0.005	1.865	0.024	0.21615	0.00078	1.865	0.024	0.21610	0.00078	2768	30	2952	6	6.2
D	12	12.1	611	37	0.06	0.009	1.642	0.013	0.26011	0.00187	1.642	0.013	0.26004	0.00187	3066	20	3247	11	5.6

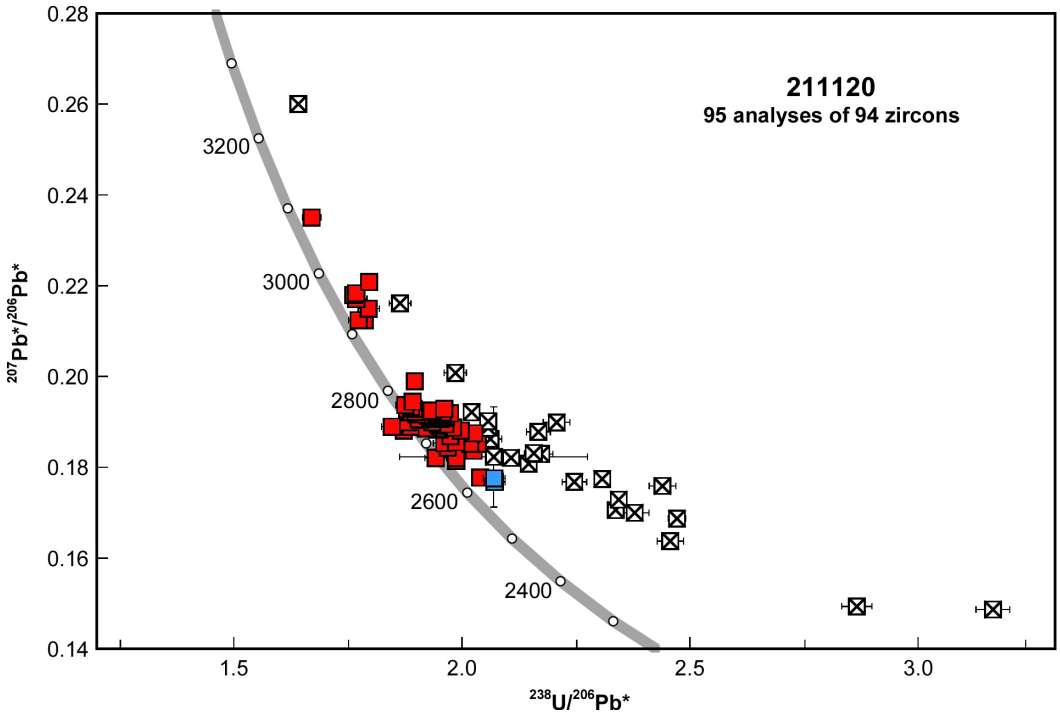

Figure 3. U–Pb analytical data for zircons from sample 211120: altered breccia, Deflector prospect. Blue squares indicate Group Y (youngest detrital zircon); red squares indicate Group S (older detrital zircons); crossed squares indicate Group D (discordance >5% or high within-run variation)

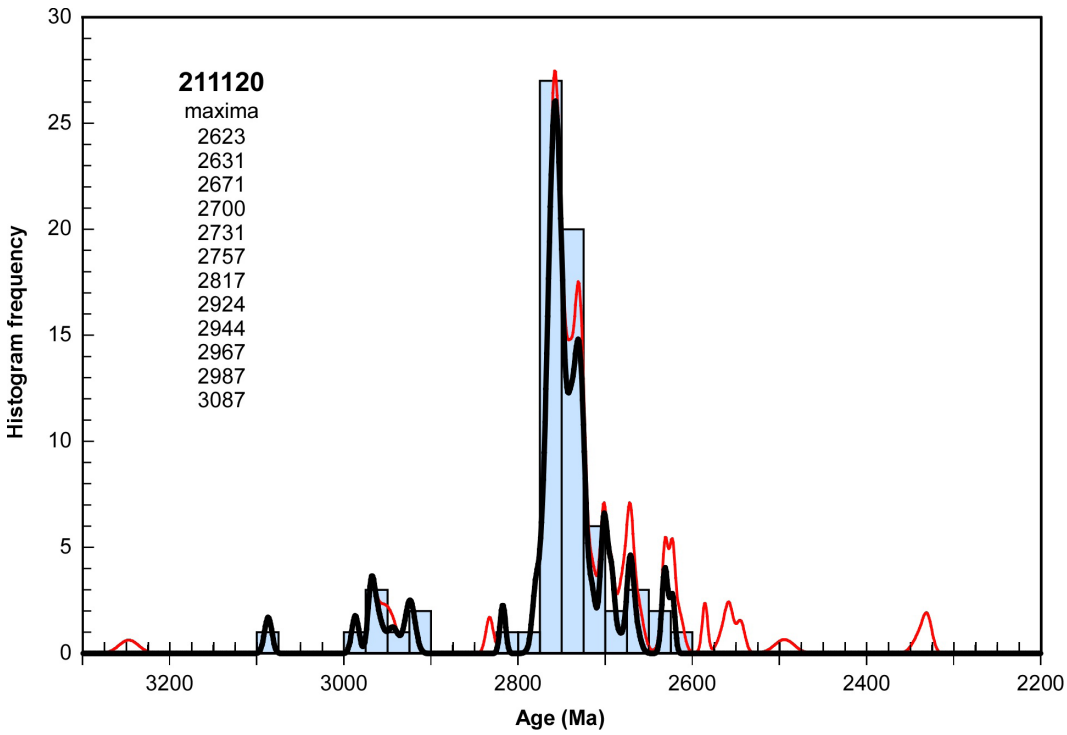

Figure 4. U–Pb analytical data for zircons from sample 211120: altered breccia, Deflector prospect. Thick curve, maxima values, and frequency histogram (bin width 25 Ma) include accepted data (71 analyses of 70 zircons); thin curve includes all data (95 analyses of 94 zircons)

Group Y comprises two analyses of one zircon (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2625 ± 5 Ma (2σ, MSWD = 2.6).

Group S comprises 69 analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 3087–2632 Ma.

It is possible that all of the analyses in Group Y and S are of unmodified detrital zircons, in which case the date of 2625 ± 5 Ma for the two analyses in Group Y represents a maximum age of deposition for the sedimentary precursor. A more conservative estimate of the maximum depositional age can be based on the weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2672 ± 10 Ma (MSWD = 1.7) for the youngest coherent group of four analyses in Group S.

The data for combined Groups Y and S indicate significant age components at c. 2967, 2757, 2731, 2700 and 2671 Ma, based on contributions from approximately 3, 28, 14, 5 and 3 analyses, respectively. These are interpreted as the ages of zircon-crystallizing rocks in the detrital source region(s), or the ages of detrital components within sediments which have been reworked into this rock

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Recommended reference for this publication

Wingate, MTD, Lu, Y, Fielding, IOH and Ivanic, TJ 2021, 211120: altered breccia, Deflector prospect; Geochronology Record 1823: Geological Survey of Western Australia, 7p.

Data obtained: 19 July 2019

Data released: 26 November 2021